Developer Technology Trends: Stack Overflow Survey Analysis

Data Analyst Capstone Project

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EXECUTIVE SUMMARY



Stack Overflow Developer Survey: 18,845 professional developers analysed

- **Languages:** JavaScript (79%), SQL (67%) and HTML/CSS (66%) lead today; they stay on top next year
- **Rising languages:** Rust and Go are the fastest-growing choices (wanted by 2.5x and 1.7x more developers than use them)
- **Databases:** PostgreSQL dominates now (61%) and next year (65%); Redis climbs to #2
- **Cloud:** AWS leads platforms (58%), ahead of Azure (36%) and Google Cloud (29%)
- **Tools:** Visual Studio Code is the most popular IDE (80%), followed by Visual Studio and IntelliJ IDEA
- **Who responded:** 69% are aged 25 to 44; 46% hold a bachelor's degree; the USA, Germany and India lead
- **Jobs and pay:** Swift, Python and C++ pay the highest average salaries (US$114K to US$131K)

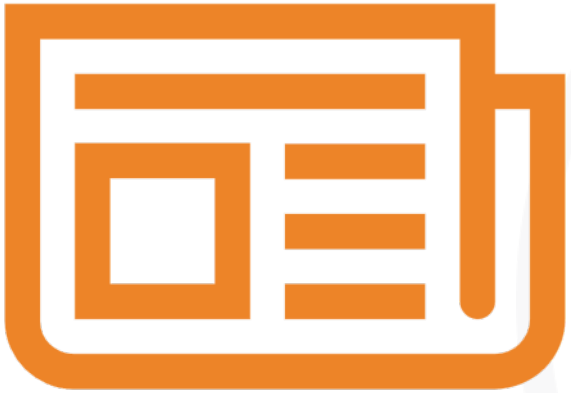
INTRODUCTION



Purpose, target audience and value of the analysis

- **Purpose:** help our IT and business consulting firm stay competitive by identifying in-demand and emerging technology skills
- **Target audience:** firm leadership, consultants, hiring managers and training teams
- **Value:** evidence-based guidance on which skills are in demand now and which are gaining momentum
- **Questions answered:**
  - Which technologies are rising or declining for next year?
  - Who are the respondents (age, country, education)?
  - Which programming languages, databases and IDEs are most in demand?

METHODOLOGY



Data Sources

- Stack Overflow Developer Survey subset (survey_data_updated.csv: 18,845 responses, 114 columns)
- Jobs API (jobs.json: 27,005 job postings)
- Programming-language salary web page (Programming_Languages.html)

Data Collection Methods

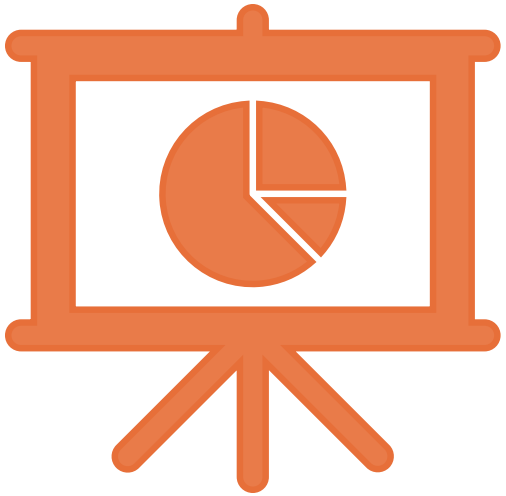
- Downloaded the survey CSV and loaded it with pandas; queried it with SQL in SQLite
- Called the Jobs REST API with Python requests and saved results to job-postings.xlsx
- Web-scraped salaries with BeautifulSoup and saved them to popular-languages.csv

Data Wrangling Steps

- Found and removed duplicate rows; confirmed ResponseId is unique
- Imputed missing values (most frequent value for categories, median for compensation)
- Normalised compensation (min-max, z-score, log) and removed outliers with the IQR method
- Split multi-select answers (e.g. "Python;SQL") into one row per technology

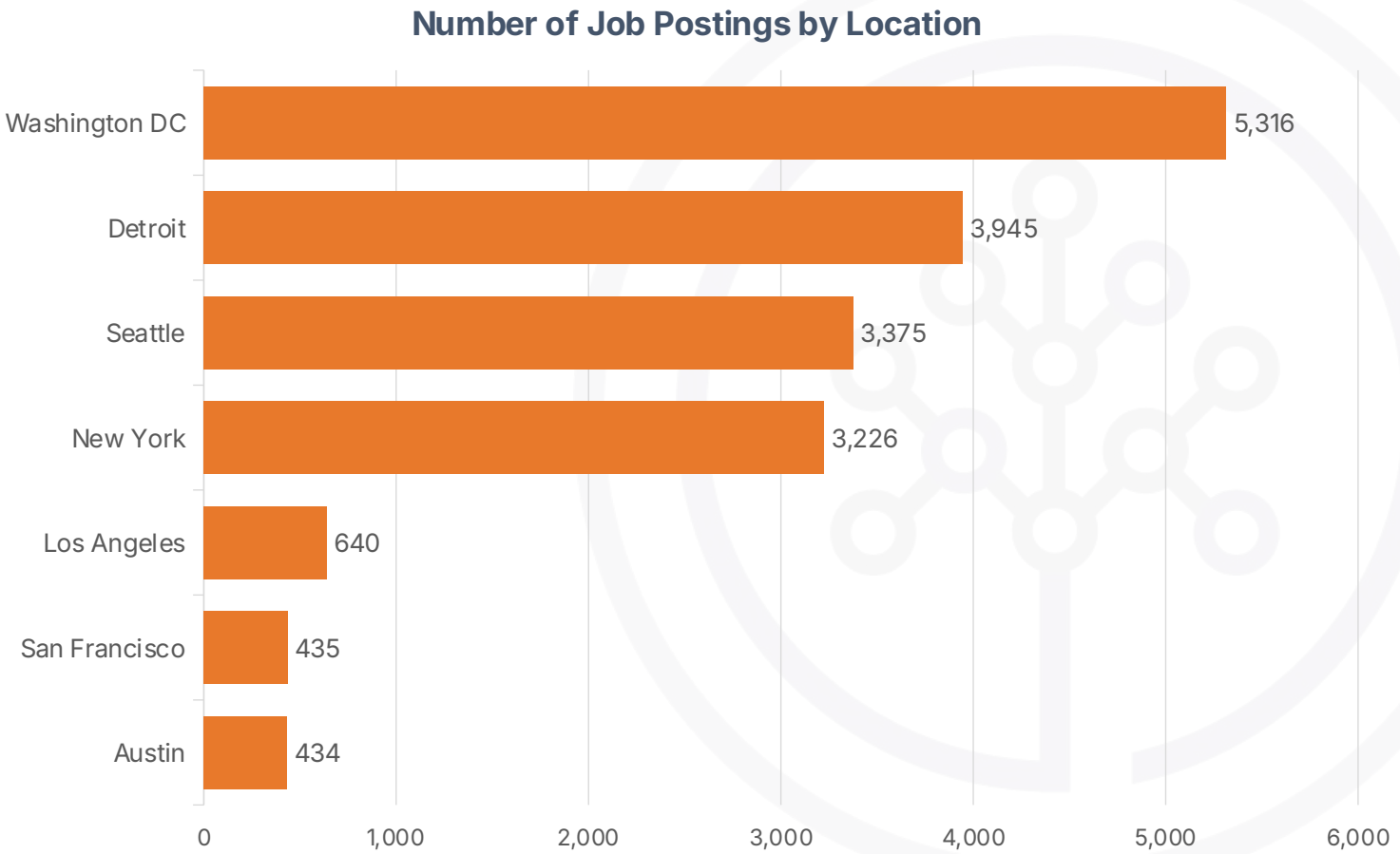
Analysis: pandas, matplotlib and seaborn for EDA; IBM Cognos Analytics for the dashboard

Results



The following slides present job postings, language salaries, and programming language and database trends.

JOB POSTINGS



Key observations

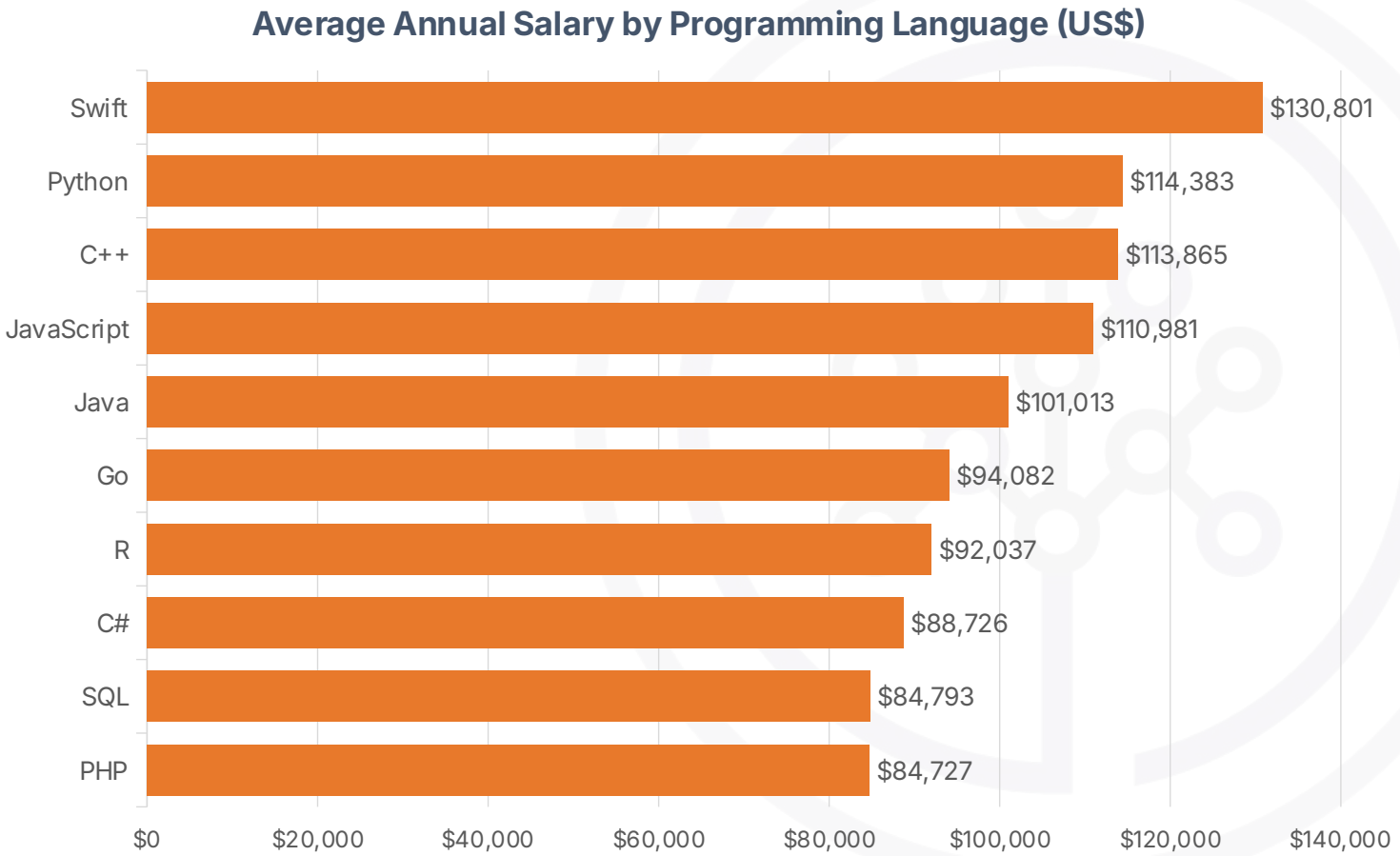
Washington DC has the most job postings (5,316)

Detroit, Seattle and New York each have more than 3,000

Los Angeles, San Francisco and Austin have far fewer (under 700 each)

Source: job-postings.xlsx (Jobs API lab, 27,005 job records)

POPULAR LANGUAGES



Key observations

Swift pays the most at $130,801 a year

Python ($114,383) and C++ ($113,865) follow closely

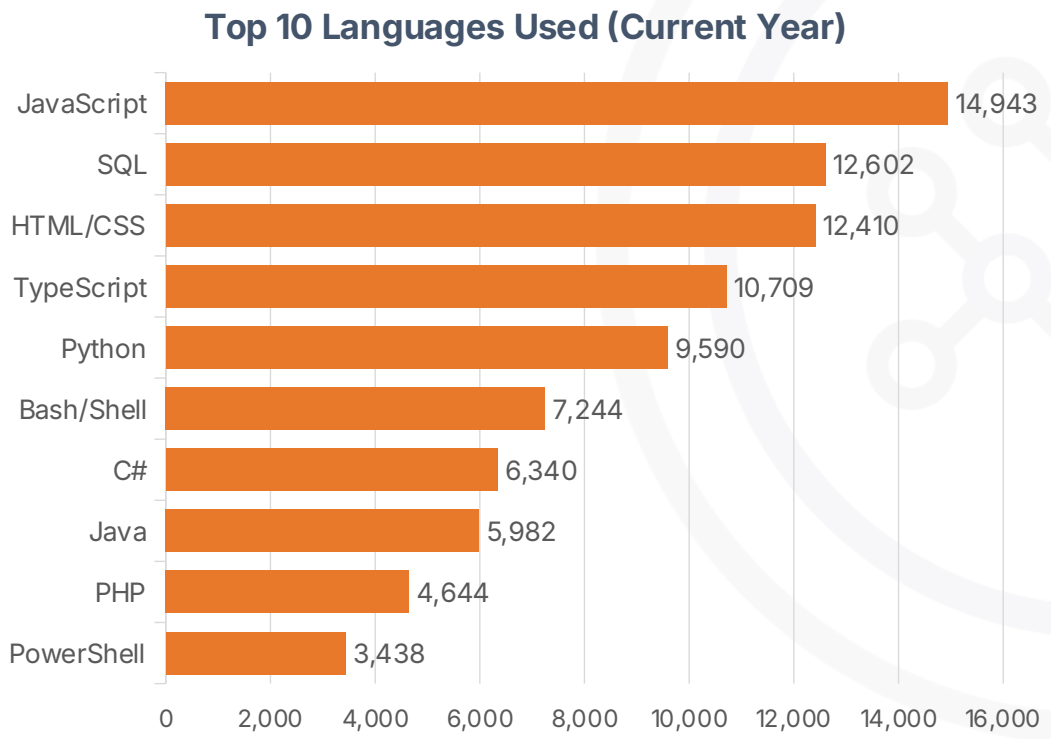
SQL and PHP are the lowest paid (about $85K) even though they are widely used

Source: popular-languages.csv (web scraping lab)

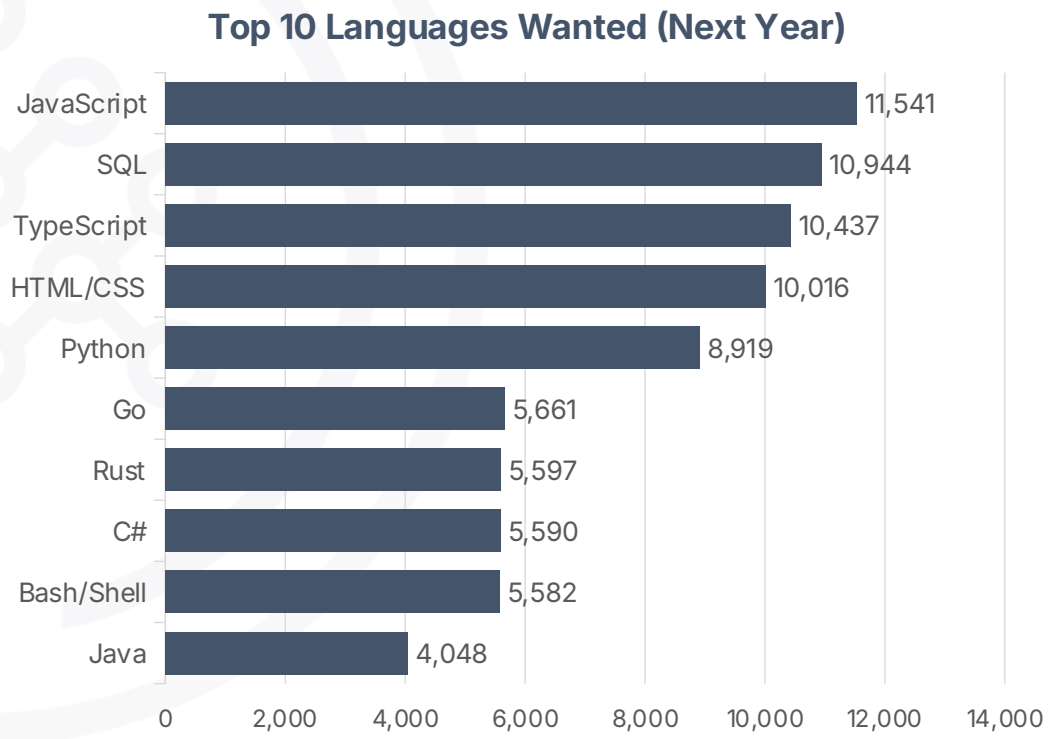
PROGRAMMING LANGUAGE TRENDS

JavaScript, SQL, HTML/CSS, TypeScript and Python lead both years. Go and Rust join the top 10 next year, replacing PHP and PowerShell.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript is the most used language (79% of respondents) and stays #1 next year (61%)
- SQL, HTML/CSS, TypeScript and Python make up the rest of the top 5 in both years
- Go and Rust enter next year's top 10; Rust is wanted by 2.5x more developers than currently use it
- PHP and PowerShell drop out of next year's top 10, and Java falls from #8 to #10

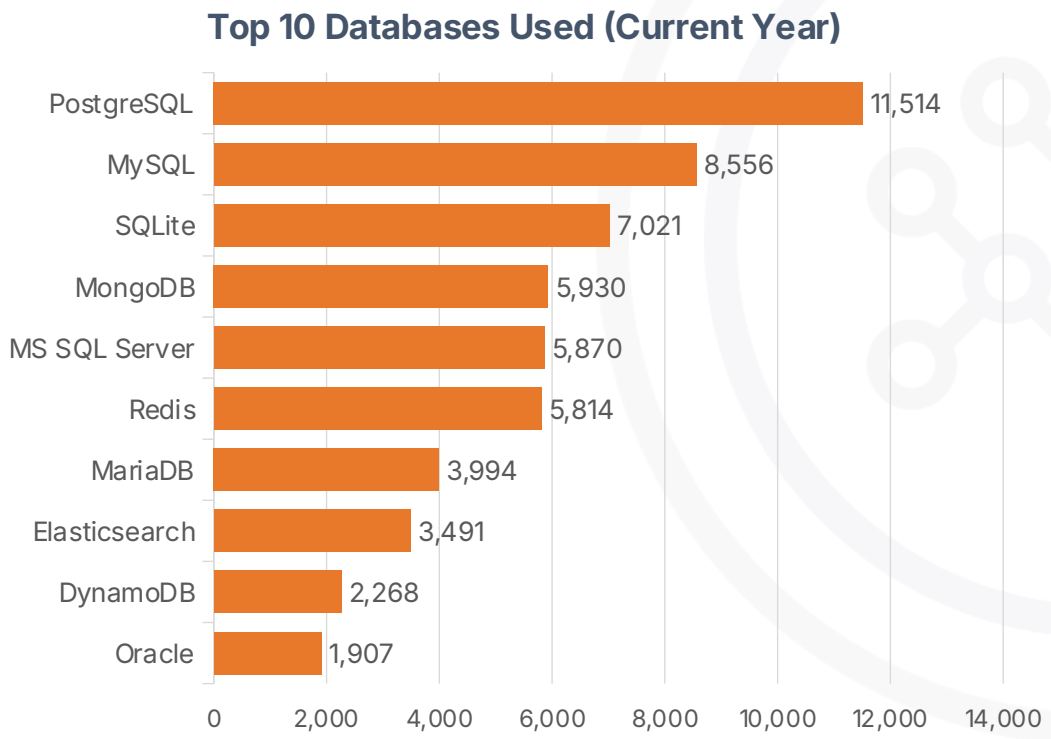
Implications

- JavaScript/TypeScript and SQL remain must-have skills for consultants and job seekers
- Steady demand for Python supports growing data, AI and automation work
- The firm should start training staff in Go and Rust to meet rising client demand
- Teams relying on PHP or legacy scripting face a shrinking talent pool and should plan modernisation

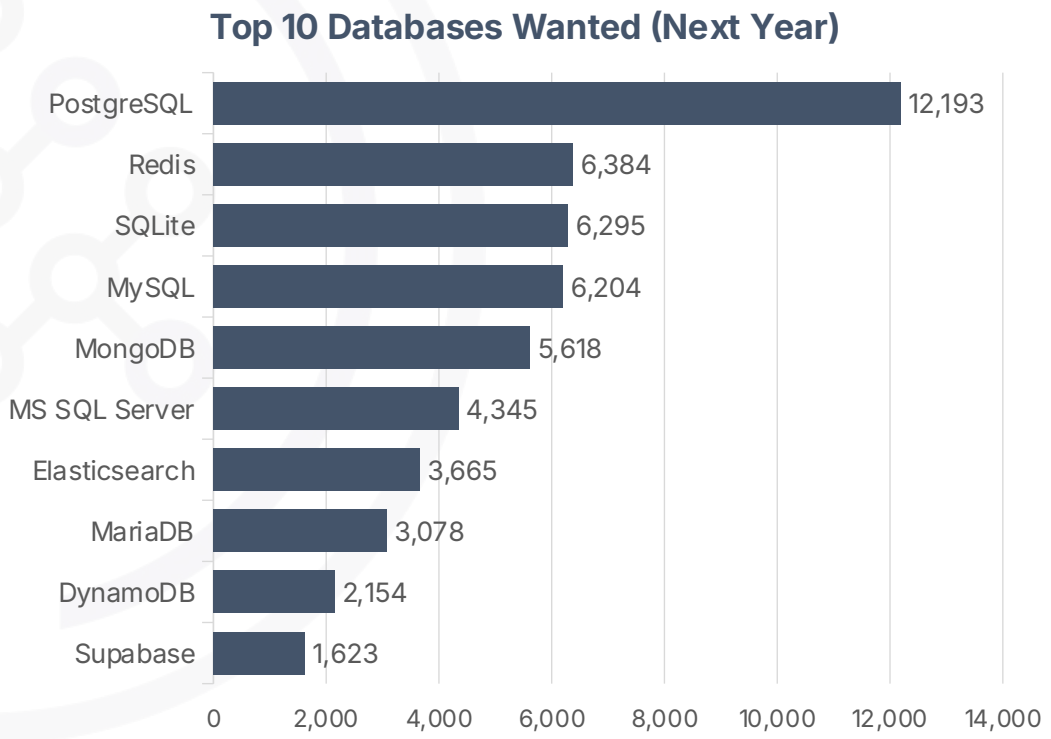
DATABASE TRENDS

PostgreSQL leads both years by a wide margin. Redis rises from #6 to #2, while MySQL and Microsoft SQL Server lose ground.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is #1 in both years: 61% use it now and 65% want to use it next year
- Redis rises from #6 to #2 next year, and Supabase enters the top 10
- MySQL drops from #2 to #4, and Microsoft SQL Server from #5 to #6
- Oracle drops out of next year's top 10

Implications

- PostgreSQL is the most valuable database skill to build or hire for
- In-memory caching (Redis) is becoming standard in modern application stacks
- Interest is shifting from proprietary databases (SQL Server, Oracle) to open-source options, lowering licence costs
- The firm can offer PostgreSQL migration and cloud database services to clients

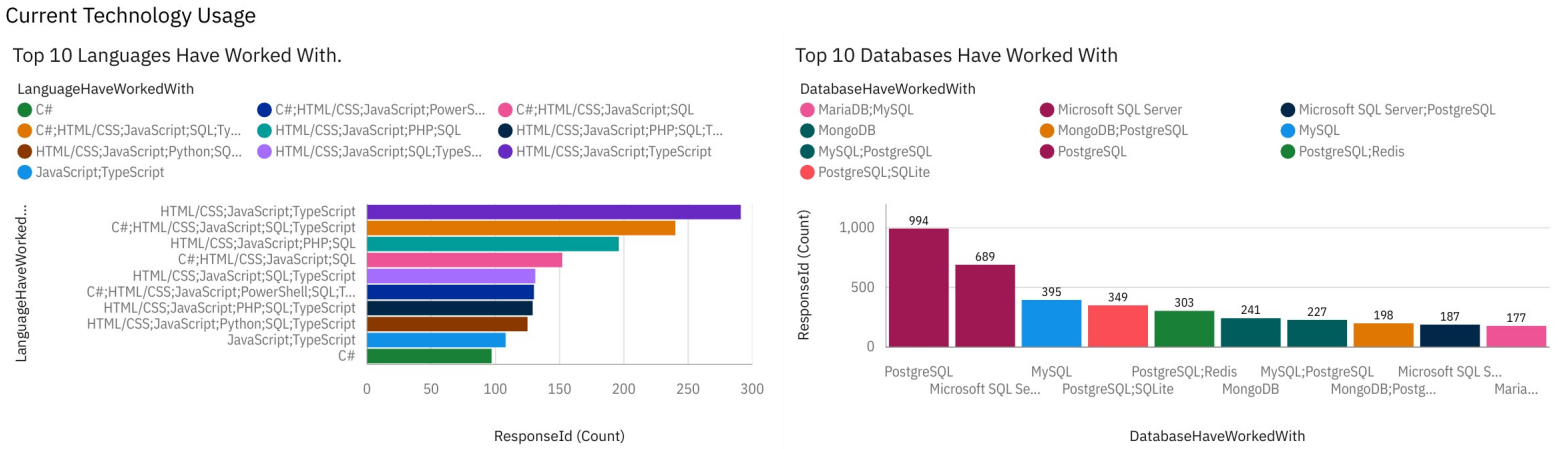
DASHBOARD

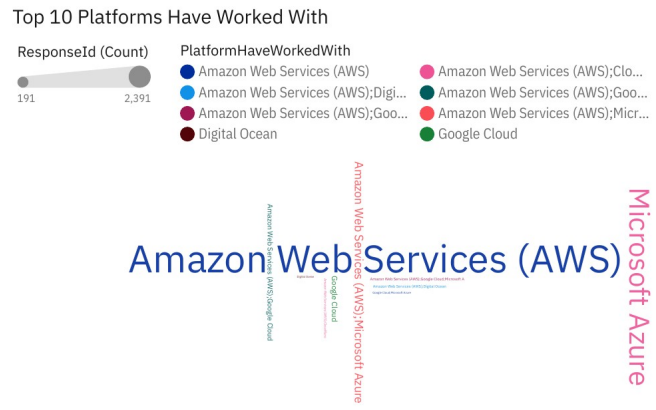


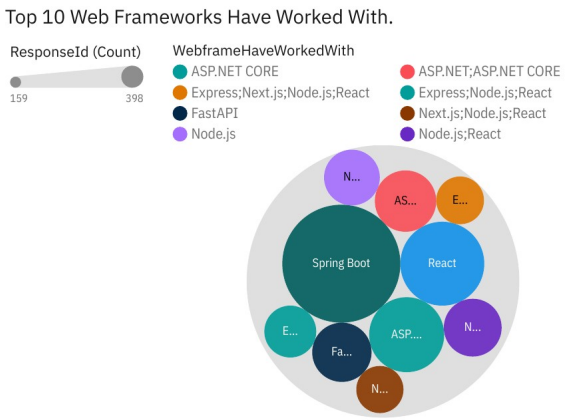
IBM Cognos Analytics dashboard: three tabs, 12 visualisations

- **Current Technology Usage:** web stacks (HTML/CSS, JavaScript, TypeScript), PostgreSQL and AWS dominate
- **Future Technology Trend:** the same leaders stay on top; interest grows in PostgreSQL, FastAPI and Phoenix
- **Demographics:** most respondents are 25 to 44, hold a bachelor's degree and live in the USA, Germany or India

DASHBOARD TAB 1: Current Technology Usage







Dashboard insight

Front-end web stacks dominate: the top language combination is HTML/CSS + JavaScript + TypeScript

PostgreSQL is the most used single database (994 respondents use only PostgreSQL)

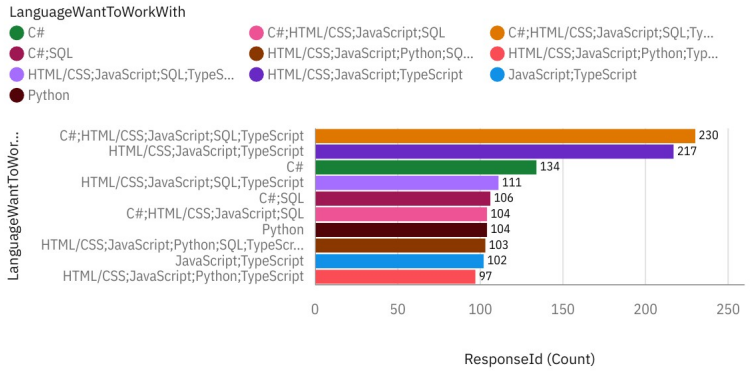
AWS is the leading cloud platform, followed by Microsoft Azure

Spring Boot, React and ASP.NET Core are the most common web framework answers

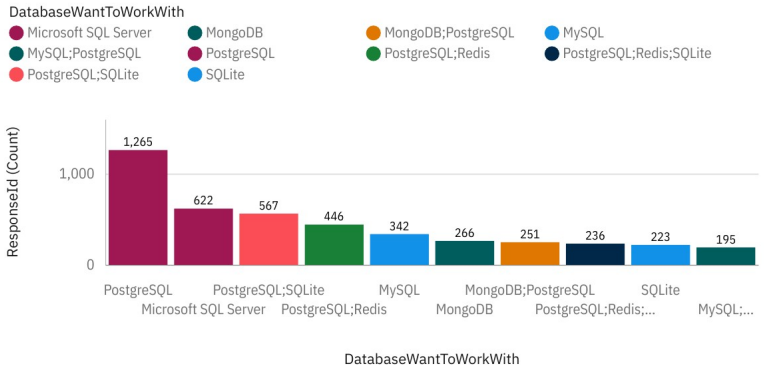
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trend

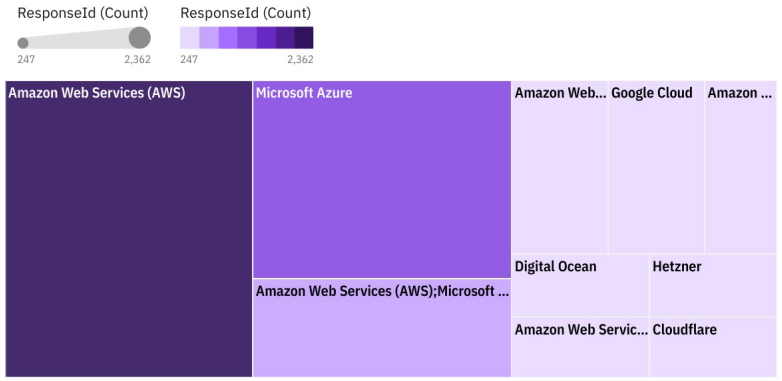
Top 10 Language Want To Work With



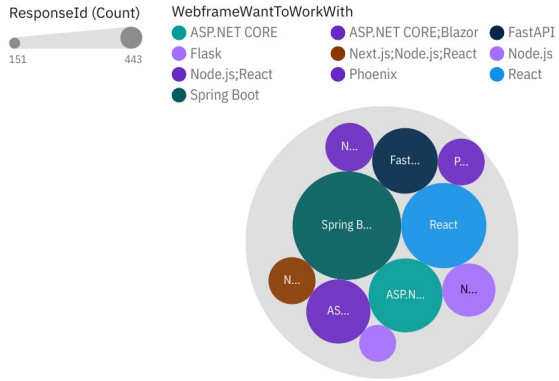
Top 10 Database Want To Work With



Top 10 Platform Want To Work With



Top 10 Webframe Want To Work With



Dashboard insight

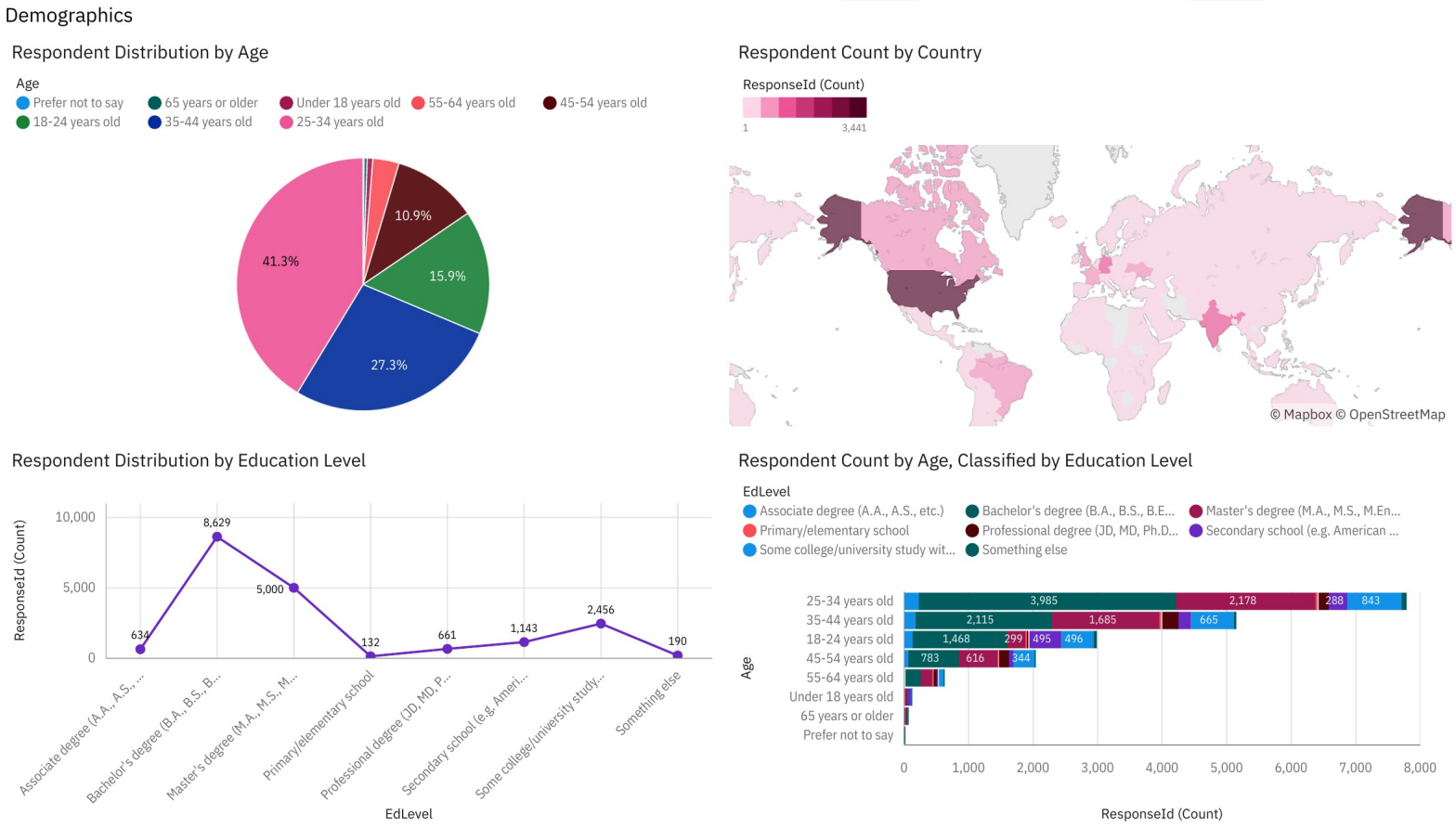
C#, HTML/CSS, JavaScript, SQL and TypeScript together is the most wanted language set

PostgreSQL interest grows: 1,265 respondents want only PostgreSQL next year

AWS and Azure remain the most wanted platforms

Spring Boot, React and ASP.NET Core stay top; FastAPI and Phoenix gain interest

DASHBOARD TAB 3: Demographics



Dashboard insight

Ages 25 to 34 make up 41% of respondents; 35 to 44 add another 27%

The USA (3,441), Germany (1,339) and India (1,316) have the most respondents

A bachelor's degree is the most common education (8,629), then a master's (5,000)

Bachelor's holders are the largest group in every main age band

DISCUSSION



Overall insights from the dashboard

- The developer community is young and highly educated: most are 25 to 44 with a degree
- Today's stack centres on JavaScript, SQL, PostgreSQL and AWS, and the same tools lead next year
- Interest is shifting toward newer, performance-focused tools such as Rust, Go, FastAPI and Redis
- Many developers combine several technologies, so multi-skill profiles are the norm
- Limitation: the data is a subset of the full survey, so results may not fully reflect the global developer community

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL and PostgreSQL are the core skills both now and next year
- Rust, Go, Redis and Supabase show the strongest growth in interest
- AWS leads cloud adoption; most respondents are aged 25 to 44 with a degree

Implications

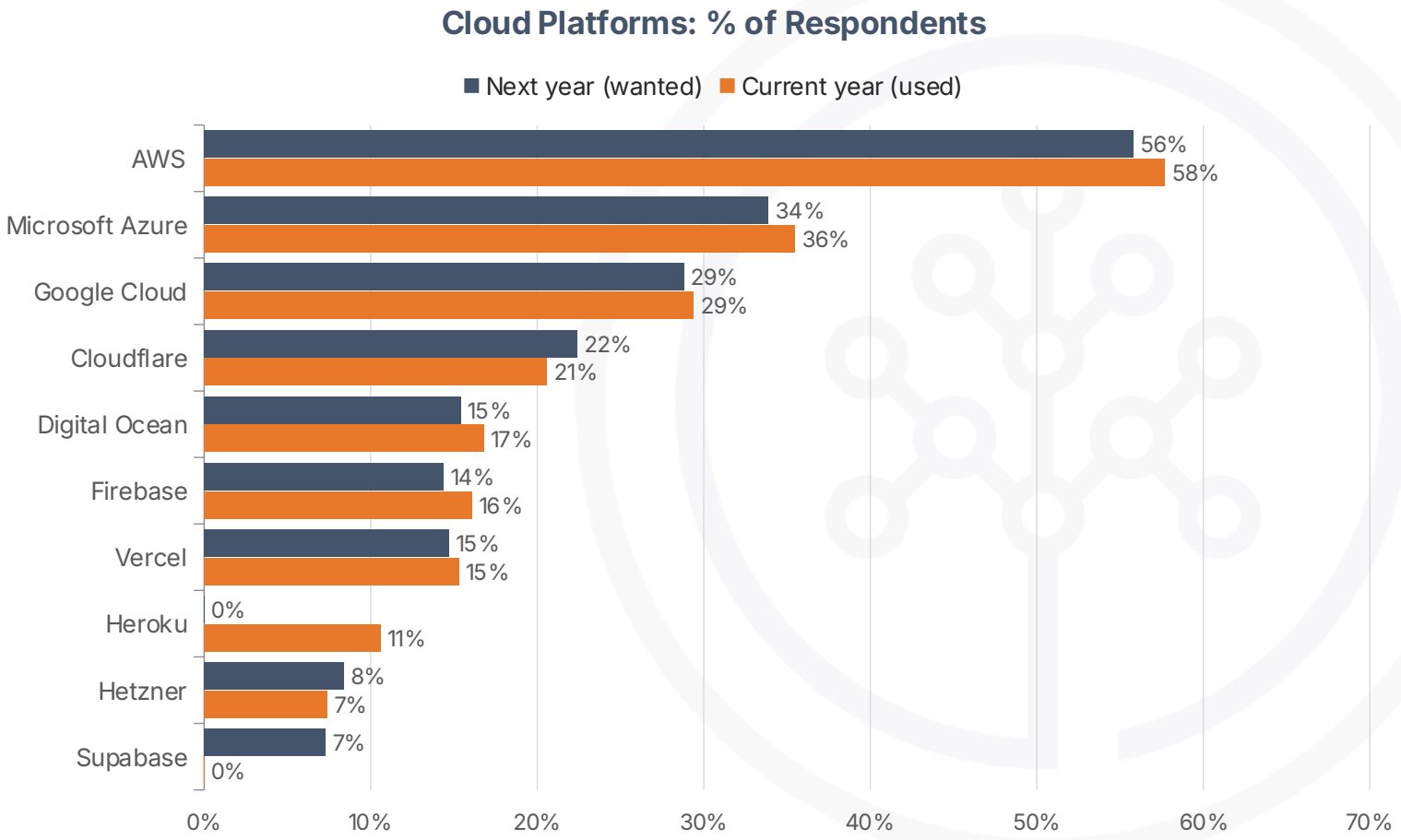
- Industry: open-source tools (PostgreSQL, Redis) and cloud platforms (AWS) are becoming the default enterprise stack
- Hiring: demand for Rust and Go may outpace supply, raising competition and pay for these skills
- Training: universities, bootcamps and employers should update curricula toward cloud, PostgreSQL and modern languages

CONCLUSION



- Web and data skills (JavaScript, TypeScript, SQL, Python) are the foundation of today's developer job market
- PostgreSQL and AWS are the leading database and cloud platform, now and next year
- Rust, Go and Redis are the technologies to watch, with interest growing faster than current use
- An interactive Cognos dashboard lets stakeholders explore technology usage, trends and demographics

APPENDIX: CLOUD PLATFORMS



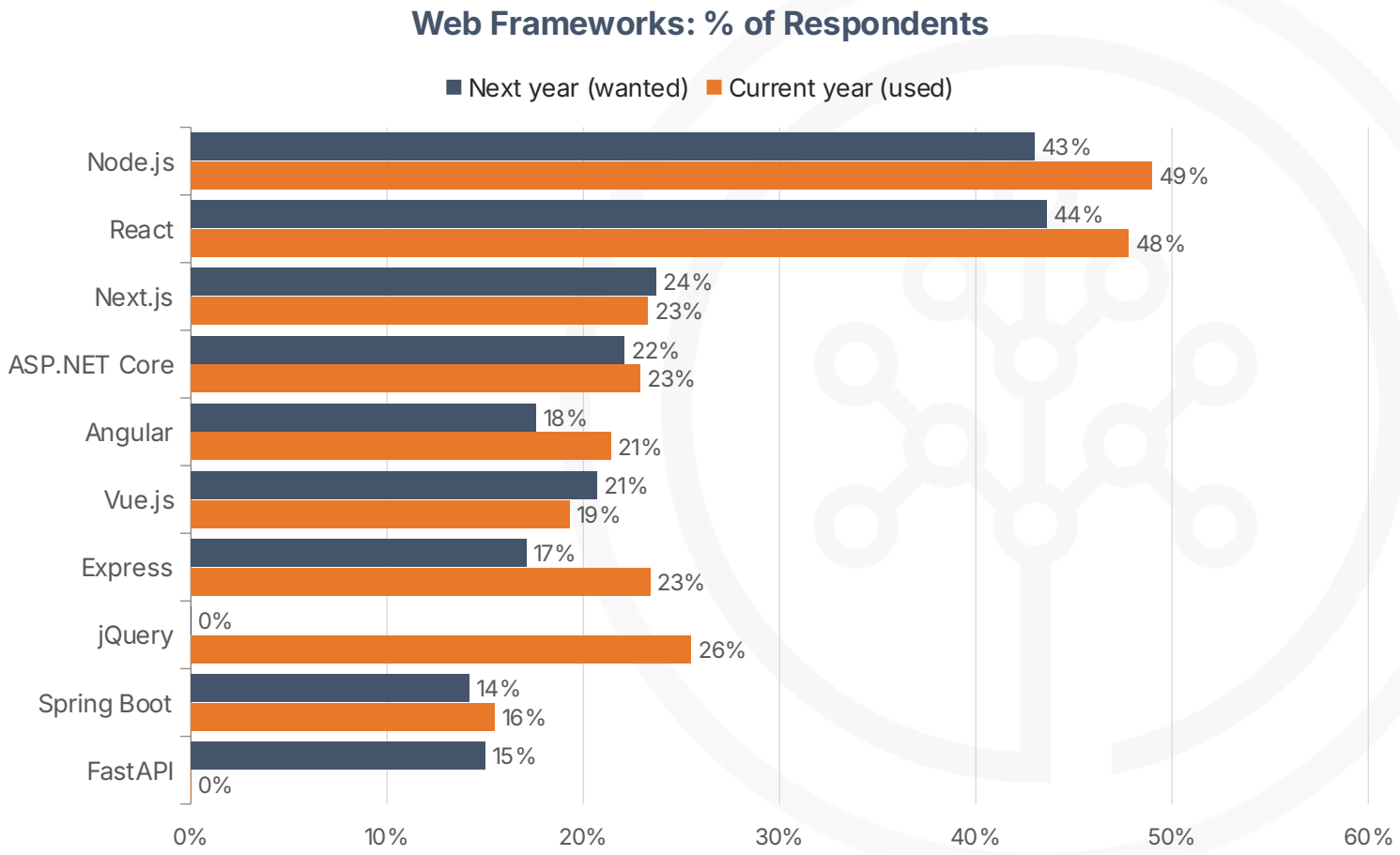
Notes

AWS leads both years (58% used, 56% wanted)

Cloudflare and Hetzner gain interest; Heroku drops out of the next-year top 10

0% = outside that year's top 10

APPENDIX: WEB FRAMEWORKS



Notes

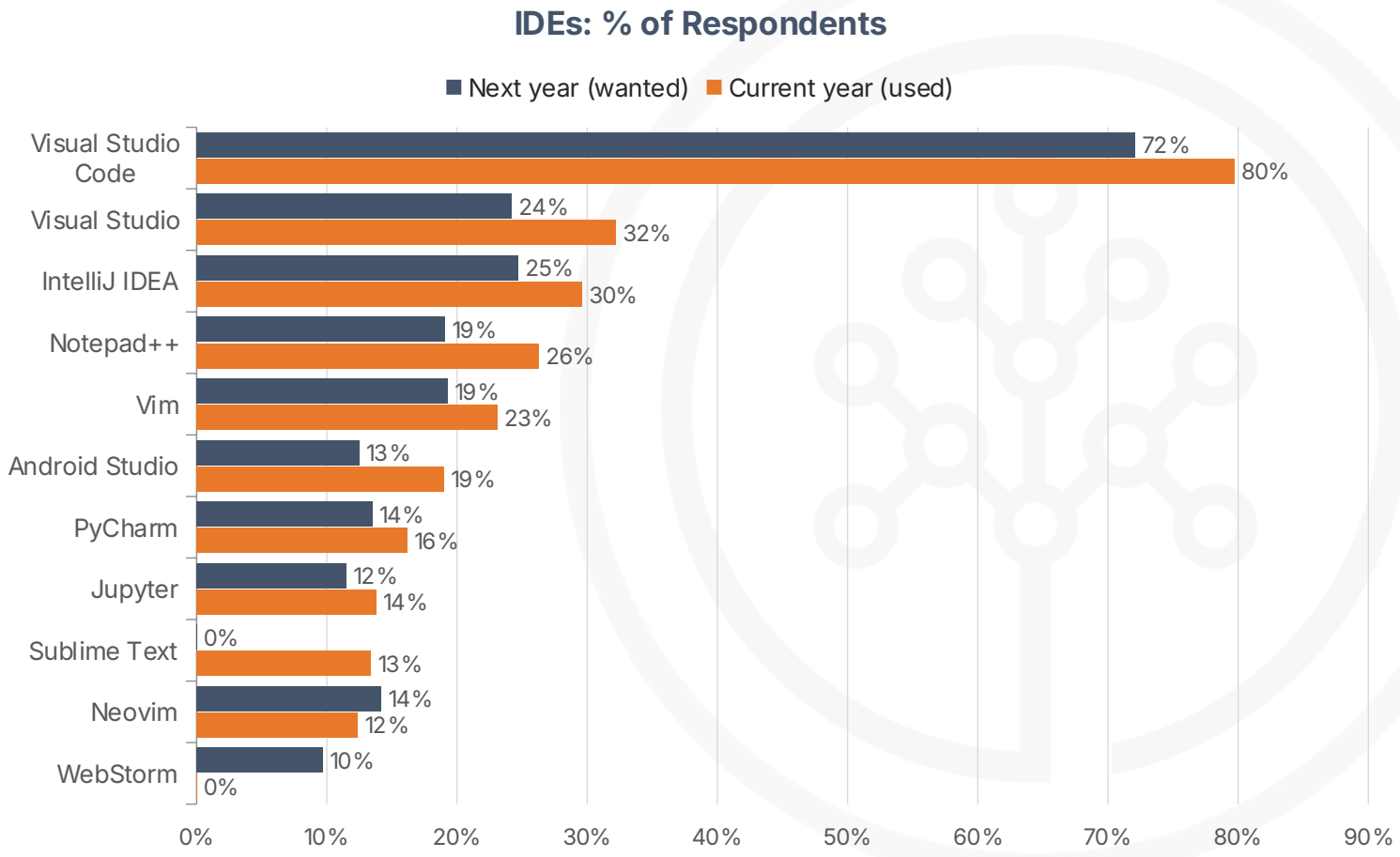
Node.js and React lead both years (about 43 to 49%)

Next.js and Vue.js gain interest; jQuery and Express decline

FastAPI enters the next-year top 10

0% = outside that year's top 10

APPENDIX: MOST POPULAR IDEs



Notes

Visual Studio Code dominates: 80% use it and 72% want to keep using it

IntelliJ IDEA overtakes Visual Studio for next year

Neovim gains interest; WebStorm enters the next-year top 10

0% = outside that year's top 10